

Here is the step-by-step solution to calculate the area of the barren field given in Assignment-2. To determine the area of a polygon defined by geographic coordinates, computers compute the area by projecting the map coordinates along a given axis and forming trapezoids. The total area of the polygon is calculated by summing the areas of the trapezoids projected from each line segment, where overlapping areas automatically subtract themselves based on the direction of the boundary.

The area of a trapezoid projected from a line segment between Point 1 and Point 2 is calculated as:

- **Projecting on Easting Axis:** $\frac{1}{2} \times (Easting_2 - Easting_1) \times (Northing_1 + Northing_2)$
- **Projecting on Northing Axis:** $\frac{1}{2} \times (Northing_2 - Northing_1) \times (Easting_1 + Easting_2)$

Below are the computations based on the provided map coordinates (WGS 1984, UTM Zone 45N) 3 :

(i) Projecting Map Coordinates Along the Easting Axis

We apply the trapezoid formula along the Easting axis for every connected line segment of the polygon from A to G, and then back to A:

- **Segment A to B:** $0.5 \times (498697.64 - 498686.20) \times (2625099.66 + 2625174.01) = 30,031,565.39$
- **Segment B to C:** $0.5 \times (498608.99 - 498697.64) \times (2625174.01 + 2625199.75) = -232,722,816.91$
- **Segment C to D:** $0.5 \times (498671.90 - 498608.99) \times (2625199.75 + 2625311.28) = 165,154,824.45$
- **Segment D to E:** $0.5 \times (498751.98 - 498671.90) \times (2625311.28 + 2625317.00) = 210,235,156.33$
- **Segment E to F:** $0.5 \times (498920.70 - 498751.98) \times (2625317.00 + 2625191.17) = 442,932,869.22$
- **Segment F to G:** $0.5 \times (498786.29 - 498920.70) \times (2625191.17 + 2625102.52) = -352,845,987.44$
- **Segment G to A:** $0.5 \times (498686.20 - 498786.29) \times (2625102.52 + 2625099.66) = -262,746,368.10$

Summing these individual segment areas gives: $30031565.39 - 232722816.91 + 165154824.45 + 210235156.33 + 442932869.22 - 352845987.44 - 262746368.10 = 39,242.94$

Total Area via Easting Projection = 39,242.95 sq. meters (taking the absolute value).

(ii) Projecting Map Coordinates Along the Northing Axis

We apply the trapezoid formula along the Northing axis for every connected line segment 3:

- **Segment A to B:** $0.5 \times (2625174.01 - 2625099.66) \times (498686.20 + 498697.64) = 37,077,744.25$
- **Segment B to C:** $0.5 \times (2625199.75 - 2625174.01) \times (498697.64 + 498608.99) = 12,835,336.33$
- **Segment C to D:** $0.5 \times (2625311.28 - 2625199.75) \times (498608.99 + 498671.90) = 55,613,368.83$
- **Segment D to E:** $0.5 \times (2625317.00 - 2625311.28) \times (498671.90 + 498751.98) = 2,852,632.30$
- **Segment E to F:** $0.5 \times (2625191.17 - 2625317.00) \times (498751.98 + 498920.70) = -62,768,576.66$
- **Segment F to G:** $0.5 \times (2625102.52 - 2625191.17) \times (498920.70 + 498786.29) = -44,223,362.33$
- **Segment G to A:** $0.5 \times (2625099.66 - 2625102.52) \times (498786.29 + 498686.20) = -1,426,385.66$ ✓

Summing these individual segment areas gives: $37077744.25 + 12835336.33 + 55613368.83 + 2852632.30 - 62768576.66 - 44223362.33 - 1426385.66 = -39,242.94$

Total Area via Northing Projection = 39,242.95 sq. meters (taking the absolute value).

Both geometric projection methods yield an identical area of **39,242.95 sq. meters** for the barren field.